

Smart Assistive Blind Stick

An Ultrasonic Sensing & Haptic Feedback System for Obstacle Detection

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Department:	Electronics & Communication Engineering	Status:	SIMULATION VERIFIED (TINKERCAD)
Domain:	Embedded Systems / Assistive Technology	Primary Hardware:	Arduino Uno R3, HC-SR04, Piezo Buzzer

1. Executive Summary

Navigational safety remains a critical challenge for visually impaired individuals. Traditional white canes provide tactile feedback for ground-level obstacles but fail to alert users about low-hanging structures or objects at waist height. This project presents a low-cost, Arduino-based Smart Assistive Blind Stick designed to detect obstacles in a 3D forward path using ultrasonic distance measurement. The system provides immediate acoustic alerts when an object enters a critical proximity zone, improving user mobility and independence.

2. System Architecture & Working Principle

The core of the system relies on ultrasonic wave reflection. The HC-SR04 ultrasonic sensor transmits a 40 kHz high-frequency sound wave burst from its transmitter transducer. When the wave collides with an obstacle, it reflects back to the receiver transducer. The microcontroller measures the total travel time (t) in microseconds using the high-precision `pulseIn()` function.

Using the speed of sound in dry air at room temperature ($v \approx 0.0343 \text{ cm}/\mu\text{s}$), the single-direction distance (d) to the target is derived using the relation:

$$d = (t \times 0.0343) / 2$$

When the computed distance falls below the defined threshold ($d < 30 \text{ cm}$), digital pin 8 sets high, actuating the audio alarm.

3. Circuit Schematic & Simulation Analysis

The circuit was prototyped and verified within Tinkercad Circuits prior to hardware deployment. The schematic establishes high stability and low current draw from the ATmega328P microcontroller.

Component Name	Arduino Connection Pin	Electrical Purpose
HC-SR04 VCC / GND	5V / GND	System Power Supply
HC-SR04 TRIG	Digital Pin 12 (Output)	Emits 10µs ultrasonic trigger pulse
HC-SR04 ECHO	Digital Pin 11 (Input)	Measures return echo pulse width
Piezo Buzzer (+) / (-)	Digital Pin 8 (Output) / GND	Provides acoustic alert signal
Current Resistor	In-series on Pin 8 (220 Ω)	Protects pin from overcurrent draw

Tinkercad Schematic Representation

[Virtual Schematic verified in Tinkercad]

Connections: HC-SR04 (Trig: D12, Echo: D11), Buzzer (Pin 8 with 220Ω protection resistor), Shared 5V/GND Rails.

4. Firmware Implementation (Arduino C++)

```
const int trigPin = 12;
const int echoPin = 11;
const int vibPin = 8;

long pingTravelTime;
float distance;

void setup() {
  pinMode(trigPin, OUTPUT);
  pinMode(echoPin, INPUT);
  pinMode(vibPin, OUTPUT);
  Serial.begin(9600);
}

void loop() {
  digitalWrite(trigPin, LOW);
  delayMicroseconds(2);

  digitalWrite(trigPin, HIGH);
  delayMicroseconds(10);
  digitalWrite(trigPin, LOW);

  // Timeout added (30,000 us) to prevent hanging
  pingTravelTime = pulseIn(echoPin, HIGH, 30000);

  if (pingTravelTime == 0) {
    distance = 400; // Object out of range
  } else {
    distance = (pingTravelTime * 0.0343) / 2.0;
  }

  Serial.print("Distance: ");
  Serial.print(distance);
  Serial.println(" cm");
}
```

```
if (distance < 30) {
  digitalWrite(vibPin, HIGH);
} else {
  digitalWrite(vibPin, LOW);
}

delay(100);
}
```

5. Physical Implementation Plan & Assembly Placeholders

The physical build transfers the verified virtual model onto an ergonomic lightweight cane. Below are the designated assembly parameters and measurement placeholders reserved for physical build verification.

Physical Assembly Notes: Component mounting requires zip-ties and hot adhesive on a PVC/aluminum cane body. Power is supplied via a 9V alkaline battery connected through a DC barrel jack.

Physical Metric / Parameter	Target Specification	Physical Prototype Measurement
Total Physical Weight	< 400 grams	[Pending Physical Build]
Battery Life (Continuous Use)	> 12 hours (9V Battery)	[Pending Physical Build]
Max Detection Range	400 cm (Theoretical)	[Pending Physical Test]
Effective Alert Threshold	30 cm (Calibrated)	[Verified in Tinkercad]
Enclosure Dimensions	Compact 3D Printed / Custom Box	[Pending Physical Build]

[Physical Hardware Prototype Photo Placeholder]

Attach high-resolution photos of the physical cane, mounting chassis, and breadboard wiring once assembled.

6. Conclusion & Future Upgrades

The simulated prototype successfully demonstrates basic obstacle detection using distance calculations and acoustic feedback. The project serves as a foundational step in embedded engineering, proving software logic and hardware integration. Future iterations will incorporate haptic vibration motors, GPS/GSM emergency alert buttons, and water detection sensors.